

Proactive Zero-Trust Intrusion Detection for Consumer IoT Applications Using Lightweight Ensemble Learning With Anomaly Analysis

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Abstract—The rapid proliferation of consumer IoT devices from smart home hubs to wearables has expanded the attack surface, introducing new security and privacy challenges. Traditional Intrusion Detection Systems (IDS) often rely on implicit trust and heavyweight computation, making them unsuitable for resource-constrained consumer electronics. This paper presents a proactive, lightweight zero-trust IDS tailored for consumer applications. Our two-layer architecture integrates a supervised stacked ensemble classifier (Random Forest, XGBoost, LightGBM) to detect known threats and an unsupervised DBSCAN (Density-Based Spatial Clustering of Applications with Noise)-based anomaly detector to identify zero-day attacks. We introduce a feature reduction pipeline driven by correlation and variance analysis, reducing the feature set by over 50% to fit edge hardware constraints. Evaluated on the CICIDS collection dataset (over 9 million flows), the framework achieves 98.48% accuracy while maintaining real-time processing capability on Raspberry Pi-class hardware. By continuously scrutinizing both malicious and benign traffic, our system delivers proactive, trust-enhancing defense critical to modern consumer IoT applications.

Index Terms—Zero-trust, intrusion detection, consumer IoT security, ensemble learning, anomaly detection, edge computing, adaptable defense.

I. INTRODUCTION

CONSUMER applications such as smart home hubs, wearable health trackers, and connected appliances have become deeply integrated into daily life. However, this hyper-connectivity also exposes consumer devices to sophisticated cyber threats. Traditional IDS solutions, typically designed for centralized infrastructures, often assume implicit trust among components and demand computational resources unsuitable for resource-constrained consumer devices [1]. In response, zero-trust security frameworks are gaining traction, treating all entities as untrusted by default and requiring continuous verification. Yet, proactively detecting known and unknown threats under stringent hardware and latency constraints remains challenging. This work presents a lightweight, two-layer zero-trust IDS designed specifically for consumer IoT applications. The system combines supervised ensemble learning for high-accuracy detection of known threats with unsupervised anomaly detection to proactively uncover zero-day attacks.

Designing effective intrusion detection for consumer IoT environments requires lightweight deployment, detection of unknown threats, and trust enforcement. The literature addressing the following domains is discussed here.

A. Zero-Trust Security in IoT

Zero-Trust Architecture (ZTA) assumes no implicit trust across devices or networks and enforces continuous verification [2], [3]. While ZTA has gained traction in enterprise contexts, its extension to consumer IoT is still emerging [4]. Alalmaie [5] implemented a Zero-Trust edge IDS, integrating access control and anomaly detection for smart city IoT, but their deep learning approach introduces latency and energy challenges.

B. Lightweight IDS for Consumer IoT

IoT-focused IDSs often prioritize low computational cost. Roy [6] proposed an IDS combining Principal Component Analysis (PCA) and Support Vector Machine (SVM) for smart homes, achieving moderate accuracy with minimal resource demands. Bafghi [7] introduced LightGBM-based

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TABLE I
COMPARISON OF EXISTING IDS APPROACHES FOR IoT SECURITY

Work	Zero-Trust	Zero-Day	Light-weight	Learning Type	Real-Time Ready	Dataset Used	Edge Deployable	IoT Focused	Remarks
[5] (2023)	✓	✗	✗	Deep Learning	✗	UNSW-NB15	✗	Smart City	High accuracy, but resource-heavy
[14] (2024)	✓	✗	✗	Federated Learning	✗	BoT-IoT	✗	Generic IoT	Needs frequent model sync
[6] (2022)	✗	✗	✓	PCA + SVM	✓	CICIDS2017	✓	Smart Home	Poor zero-day detection
[11] (2023)	✗	✓	✗	RL + Autoencoder	✗	NSL-KDD	✗	Industrial IoT	Accurate but heavy
[7] (2021)	✗	✗	✓	LightGBM	✓	KDD99	✓	IoT	No anomaly generalization
[12] (2020)	✗	✓	✓	DBSCAN	✓	IoT Botnet	✓	IoT Botnet	Unsupervised only
[10] (2023)	✗	✗	✗	Ensemble (Bagging)	✗	NSL-KDD	✗	General	High accuracy, but costly
[15] (1996)	✗	✓	✓	Clustering	✗	-	Algorithmic	Generic	Foundational DBSCAN method
Proposed	✓	✓	✓	Ensemble + DBSCAN	✓	CICIDS2017	✓	Consumer IoT	ZT-layered detection with low overhead

lightweight IDS with efficient training but limited generalization to unknown [8]. These solutions either lack anomaly detection or do not integrate proactive ZT policies [9].

C. Ensemble and Hybrid Learning Approaches

Nassreddine [10] evaluated ensemble classifiers for network attacks, showing robustness but requiring powerful nodes. Chen [11] adopted a hybrid deep RL and autoencoder model to capture temporal dependencies, but their design suits cloud settings, not edge devices.

D. Unsupervised Anomaly Detection in IoT

Unsupervised models like DBSCAN are valuable for zero-day detection. Alsmadi and Xu [12] demonstrated DBSCAN's utility for IoT botnet detection. However, unsupervised models alone often misclassify benign outliers as [4], [13]. Our work combines the strengths of supervised ensemble learning and unsupervised DBSCAN in a two-layer ZT framework optimized for consumer devices.

Table I compares selected works across key dimensions. As seen, few systems simultaneously enforce zero-trust, support lightweight edge deployment, and detect novel threats. Our approach addresses this triad by integrating supervised ensemble classifiers with unsupervised clustering in a layered zero-trust design suitable for consumer applications.

II. PROBLEM STATEMENT AND MOTIVATION

The proliferation of consumer edge- IoT devices has reshaped the digital landscape, embedding computation into daily-use electronics such as smart thermostats, security cameras, wearable devices, and home automation systems using IoT applications and protocols [16], [17]. Despite their growing ubiquity, these devices often remain highly vulnerable to cyberattacks due to their inherent design constraints and lack of robust, integrated security frameworks. Consumer IoT environments pose distinct challenges:

- **Resource Constraints:** Most devices operate on low-power processors with limited RAM and storage, severely restricting the use of conventional or compute-intensive security solutions [18].
- **Heterogeneity and Dynamism:** IoT networks are dynamic, composed of a wide variety of protocols, device types, and transient connectivity states-complicating the design of generalized intrusion detection [19], [20].
- **High Privacy Sensitivity:** Consumer data, including voice, video, health, and location information, necessitates privacy-preserving models with minimal data leakage and low tolerance for false outcome [21], [22].

TABLE II

SYMBOL TABLE: NOTATION AND DESCRIPTION

Symbol	Description
$x \in \mathbb{R}^d$	Input flow with d -dimensional features before reduction
$x' \in \mathbb{R}^k$	Reduced feature vector after feature selection, $k \ll d$
g	Meta-classifier combining ensemble outputs
h_1, h_2, h_3	Base classifiers: Random Forest, XGBoost, LightGBM
$p_{ens}(x')$	Aggregated ensemble probability score for input x'
θ_{ens}	Decision threshold for known attack classification (Layer 1)
$\mathcal{A}, \varepsilon$	Anomaly detector, DBSCAN neighborhood radius
$\min Pts$	Minimum neighbors to form a DBSCAN core point
$N_\varepsilon(x'), C_i$	ε -neighborhood of x' , Cluster i identified in DBSCAN
$\mathcal{O}$	Set of outliers (noise points) in DBSCAN clustering
T_E	Inference latency of ensemble classifier
T	Average per-flow system inference latency
M	Peak RAM usage during inference
S	Total model size in storage (MB)
$\mathcal{D}_{known}, \mathcal{D}_{zero-day}$	Known attack, zero-day attack distribution
$\mathcal{D}_{benign}$	Truly benign sample distribution
$\mathcal{A}_{benign}$	Samples classified as benign by Layer 1
$\mathcal{O}_{TP}, \mathcal{O}_{FP}$	True +ve and False +ve anomalies as DBSCAN outliers
τ_1, τ_2	Accuracy/anomaly precision thresholds
γ, ρ	Maximum allowable false +ve rate in L2, Noise Ratio in L2
AUC	Area Under ROC Curve for Layer 1 performance

Conventional IDS frameworks either rely heavily on centralization or operate under static trust assumptions, rendering them inadequate for zero-day resilience and continuous verification in evolving consumer IoT topologies. Additionally, models optimized for enterprise-scale networks are computationally impractical for constrained platforms Raspberry Pi, and can be implemented using ESP32, or similar microcontroller-based systems. To address these gaps, our motivation stems from the following core objectives:

- **Proactive Zero-Trust Enforcement:** Design an intrusion detection framework grounded in zero-trust architecture principles, where no entity (internal or external) is trusted by default (all communication is continuously verified).
- **Dual-Mode Detection Capability:** Integrate both supervised and unsupervised techniques to detect known (signature-based) and zero-day (anomaly-based) attacks using a layered ensemble structure.
- **Edge-Compatible Efficiency:** Ensure the proposed model maintains a lightweight computational footprint, enabling real-time inference and low power consumption on consumer-grade embedded platforms.

This research is driven by the growing need to operationalize zero-trust principles in the context of consumer IoT, where attack surfaces are wide, defense capabilities are fragmented, and user tolerance for false alerts is minimal. The goal is to build a trustworthy, scalable, and proactive defense mechanism tailored to the realities of consumer device security.

III. PROPOSED FRAMEWORK

A. Proposed Architecture: Layered Design

To satisfy the stringent security, efficiency, and scalability demands of consumer IoT applications, we propose a layered

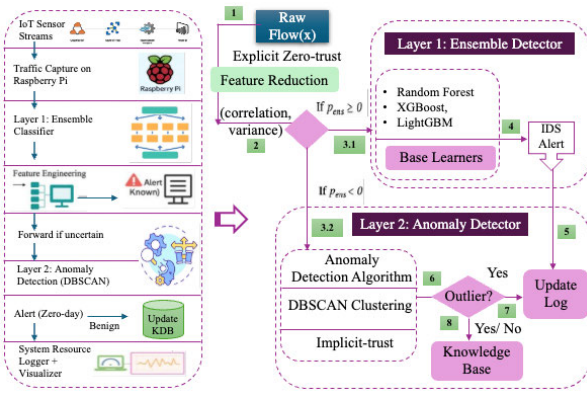


Fig. 1. Layered architecture of the proposed proactive zero-trust IDS for consumer IoT applications.

Zero-Trust Intrusion Detection System (ZT-IDS) designed for proactive threat mitigation. The architecture, illustrated in Fig. 1, follows the foundational zero-trust paradigm of continuous verification and operates in two tightly coupled analytical layers. This modular design is optimized to execute on resource-constrained edge devices such as Raspberry Pi, smart home hubs, or IoT gateways, where memory and processing power are at a premium.

1) *Feature Reduction Module*: Incoming traffic flows are first preprocessed and passed through a lightweight feature engineering unit that employs correlation-based filtering and variance thresholding. Redundant or low-informative features are discarded, reducing dimensionality while preserving discriminatory power. This module ensures the downstream classifiers receive a compact representation $x' \in \mathbb{R}^k$, where $k \ll d$, the original feature space dimension.

2) *Layer 1 – Ensemble-Based Classifier*: The first detection layer is a supervised ensemble meta-classifier that identifies known attack patterns employing a stacked ensemble of three heterogeneous learners: Random Forest (RF), XGBoost (XGB), and LightGBM (LGB). Each base model produces a probabilistic prediction, which is aggregated using a soft voting mechanism or a learned meta-layer:

$$p_{ens}(x') = \frac{1}{3} (f_{RF}(x') + f_{XGB}(x') + f_{LGB}(x'))$$

If the ensemble confidence exceeds a classification threshold θ , the flow is labeled as malicious and appropriate alerts are generated. Otherwise, it is forwarded to the anomaly detection layer for deeper scrutiny.

3) *Layer 2 – DBSCAN-based Anomaly Detector*: Traffic instances deemed benign by Layer 1 are subjected to unsupervised clustering using DBSCAN. This layer targets the detection of zero-day or previously unseen attacks by isolating outliers in high-dimensional space. Formally, for each point x' :

$$N_\epsilon(x') = \{x_j \in \mathbb{R}^k : \|x' - x_j\| \leq \epsilon\}$$

If $|N_\epsilon(x')| < \text{minPts}$, the sample is treated as an anomaly and flagged as a potential zero-day threat. Otherwise, it is logged as legitimate traffic.

This dual-layer structure enforces a proactive defense cycle: Layer 1 detects known threats with high precision, while Layer 2 continuously monitors for emergent anomalies

achieving both horizontal and vertical trust enforcement. By avoiding centralized inference or deep learning models, the architecture is both edge-compliant and latency-aware. Feature reduction was carried out using correlation filtering and variance thresholding, while DBSCAN parameters (ϵ , minPts) were empirically tuned via grid search to balance anomaly precision and false positive rate.

B. Research Problem Formulation and Hypotheses

This work is guided by two core research questions, each linked to a hypothesis that is evaluated through the proposed layered intrusion detection framework.

RQ1: Can a lightweight ensemble model accurately classify known attacks in consumer IoT traffic while maintaining edge deployability?

1) *Hypothesis H_1* : A compact ensemble classifier $\mathcal{E} = \{f_{RF}, f_{XGB}, f_{LGB}\}$ with dimensionality-reduced input $x' \in \mathbb{R}^k$ can achieve classification accuracy $A \geq \tau_1$ (e.g., 95%) on known attack instances with total inference time $T_{\mathcal{E}} < \delta$ ms on edge hardware. Formally,

$$\mathbb{E}_{x \sim \mathcal{D}_{\text{known}}} [\mathbb{I}(\arg \max \mathcal{E}(x') = y)] \geq \tau_1, \quad T_{\mathcal{E}} < \delta \quad (1)$$

where $\mathcal{D}_{\text{known}}$ is the known-attack traffic distribution, and $\mathbb{I}(\cdot)$ is the indicator function.

RQ2: Can DBSCAN identify previously unseen (zero-day) attacks with minimal false positives when applied only to Layer-1-benign flows?

2) *Hypothesis H_2* : Applying DBSCAN on a filtered set of benign-labeled samples X_{benign} can detect novel attacks with anomaly precision $P \geq \tau_2$ (e.g., 90%) and false-positive rate $FPR < \gamma$. Let:

$$\text{DBSCAN}_\epsilon(X_{\text{benign}}) = \mathcal{C}_1, \dots, \mathcal{C}_n, \mathcal{O} \quad (2)$$

where $\mathcal{O}$ represents outlier points, then:

$$\frac{|\mathcal{O} \cap \mathcal{D}_{\text{zero-day}}|}{|\mathcal{O}|} \geq \tau_2, \quad \text{FPR} = \frac{|\mathcal{O} \cap \mathcal{D}_{\text{benign}}|}{|\mathcal{D}_{\text{benign}}|} < \gamma \quad (3)$$

where $\mathcal{D}_{\text{zero-day}}$ and $\mathcal{D}_{\text{benign}}$ denote zero-day and truly benign instances, respectively.

C. Methodology: Mathematical Formulation and Sequence

Let $x \in \mathbb{R}^d$ be an input feature vector after reduction.

1) *Layer 1: Stacked Ensemble*: Each base classifier $h_m(x)$ maps x to a probability of being malicious: $p_m = h_m(x)$, $m \in \{1, \dots, M\}$. The ensemble combines these using a meta-classifier g (logistic regression): $p_{ens}(x) = g(h_1(x), h_2(x), \dots, h_M(x))$. Traffic is flagged malicious if $p_{ens}(x) \geq \theta_{ens}$. While the computational complexity of Random Forest is $O(n \log n)$ per tree, XGBoost / LightGBM needs $O(Kd \log n)$, with K boosting iterations.

2) *Layer 2: DBSCAN-Based Anomaly Detector*: Remaining benign traffic passes to DBSCAN. For sample pairs (x_i, x_j) , $d(x_i, x_j) \leq \epsilon$. Samples with fewer than minPts neighbors become outliers and are reclassified as malicious. The complexity is $O(n^2)$ (optimized to $O(n \log n)$ with indexing).

3) *Knowledge Base and Continuous Verification*: The IDS continuously updates the statistical knowledge base, i.e., mean and variance of benign flow as well as the cluster structure from DBSCAN, enabling adaptive detection over time.

D. Dbscan Clustering

Let $\mathcal{X} = \{x_1, x_2, \dots, x_n\} \subset \mathbb{R}^d$ be the set of benign-labeled vectors. For each point x_i , define:

$$N_\varepsilon(x_i) = \{x_j \in \mathcal{X} : \|x_i - x_j\| \leq \varepsilon\}$$

Points with $|N_\varepsilon(x_i)| < \text{minPts}$ are labeled outliers. The hyper-parameter (ε and minPts) tuning is performed by assessing the quality of the cluster using the Silhouette Score.

E. Evaluation Metric Formulation

To comprehensively assess the performance of the proposed Zero-Trust Intrusion Detection System (ZT-IDS), we formulate a multi-perspective evaluation model composed of classification accuracy, anomaly precision, false positive rate, and resource efficiency.

Layer 1: Ensemble Classifier Metrics are considered to be the Accuracy, Precision, Recall, F1-score, and Area Under the Receiver Operating Characteristic Curve (AUC-ROC) [23].

Let the set of test samples be $\mathcal{X} = \{x_i\}_{i=1}^N$ with true labels $Y = \{y_i\}_{i=1}^N$ and predicted labels $\hat{Y} = \{\hat{y}_i\}_{i=1}^N$ from the ensemble classifier $\mathcal{E}$. Then, we define:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (4)$$

$$\text{Precision} = \frac{TP}{TP + FP} \quad (5)$$

$$\text{Recall (TPR)} = \frac{TP}{TP + FN} \quad (6)$$

$$\text{F1-score} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}} \quad (7)$$

$$\text{AUC-ROC} = \int_0^1 TPR(FPR^{-1}(x)) dx \quad (8)$$

where TP , TN , FP , and FN represent true positives, true negatives, false positives, and false negatives, respectively [23].

Layer 2: Anomaly Detection Metrics (DBSCAN) Let $X_{\text{benign}} \subset \mathcal{X}$ be the samples labeled as benign by Layer 1. DBSCAN partitions these into K clusters and a set of outliers $\mathcal{O}$:

$$\text{DBSCAN}_\varepsilon(X_{\text{benign}}) \Rightarrow \{\mathcal{C}_1, \dots, \mathcal{C}_K\}, \quad \mathcal{O} = \text{Noise} \quad (9)$$

$\mathcal{O}_{\text{TP}}$: true zero-day anomalies correctly marked as outliers and $\mathcal{O}_{\text{FP}}$: benign points wrongly marked as outliers.

$$\text{Anomaly Precision} = \frac{|\mathcal{O}_{\text{TP}}|}{|\mathcal{O}_{\text{TP}}| + |\mathcal{O}_{\text{FP}}|} \quad (10)$$

$$\text{False Positive Rate}_{\text{Layer 2}} = \frac{|\mathcal{O}_{\text{FP}}|}{|\mathcal{X}_{\text{benign}}|} \quad (11)$$

Additionally, we define the Noise Ratio as:

$$\rho = \frac{|\mathcal{O}|}{|\mathcal{X}_{\text{benign}}|} \quad (12)$$

F. Algorithm: Two-Layered Proactive Zero-Trust IDS

Algorithm 1 summarizes the layered intrusion detection pipeline integrating both known and zero-day attack detection. Formally, Layer-1 ensemble classification is defined as $p_{\text{ens}}(x) = \frac{1}{3}(f_{\text{RF}}(x) + f_{\text{XGB}}(x) + f_{\text{LGB}}(x))$, and Layer-2

TABLE III

EVALUATION OF LAYER-1 AND LAYER-2 FOR DIFFERENT IOT SCENARIOS

Scenario	Accuracy (L1)	F1-Score (L1)	DBSCAN NR (L2)	Precision (L2)	DT Acc.	SVM Acc.
Smart Bulb	1.00	0.00	N/A	N/A	0.94	0.92
Spoofed Door	0.998	0.80	N/A	N/A	0.91	0.90
DNS Tunnel	0.96	0.95	N/A	N/A	0.88	0.86
Botnet Infection	0.98	0.94	N/A	N/A	0.90	0.89
Zero-Day	0.93	0.89	0.11	0.91	0.85	0.83

anomaly detection via DBSCAN as $N_\varepsilon(x) = \{x_j : \|x - x_j\| \leq \varepsilon\}$, $|N_\varepsilon(x)| < \text{minPts} \Rightarrow x \in \mathcal{O}$.

DBSCAN was selected over alternatives such as Isolation Forest and k-means because it does not assume linear separability or fixed cluster counts, making it more suitable for sparse IoT traffic where attack flows appear as density outliers. As shown in Table III, our ensemble consistently outperforms lightweight baselines such as Decision Tree and SVM across all scenarios, justifying its selection for Layer-1.

Algorithm 1 Two-Layered Proactive Zero-Trust IDS

Require: Input network flow x , lightweight ensemble classifier g , anomaly detector $\mathcal{A}$ (e.g., DBSCAN), thresholds $\theta_{\text{ens}}, \theta_{\text{an}}$

- 1: $x' \leftarrow \text{FEATUREREDUCTION}(x)$ {Preprocessing and normalization}
- 2: **Layer-1:** $p_{\text{ens}} \leftarrow g(h_1(x'), h_2(x'), h_3(x'))$ {Ensemble of base learners}
- 3: **if** $p_{\text{ens}} \geq \theta_{\text{ens}}$ **then**
- 4: **return Alert: Known Attack (L1)**
- 5: **else**
- 6: **Layer-2:** $\text{label} \leftarrow \mathcal{A}.\text{predict}(x')$ {Anomaly-based zero-day detection}
- 7: **if** $\text{label} == \text{outlier}$ **then**
- 8: **return Alert: Zero-Day Detected (L2)**
- 9: **else**
- 10: $\text{LOGFLOW}(x', p_{\text{ens}})$ {Benign flow; update edge knowledge base}
- 11: **end if**
- 12: **end if**

1) System-Level Metrics: To ensure feasibility on consumer devices, we evaluate:

Inference Latency T

$$= \frac{1}{N} \sum_{i=1}^N t_i \quad (\text{per flow}) \quad (13)$$

Memory Footprint M

$$= \text{Peak RAM usage during inference} \quad (14)$$

ModelSize S = Disk space-trained model components (15)

An efficient edge-compatible model must satisfy $T < \delta_t$, $M < \delta_m$, $S < \delta_s$, where δ_t , δ_m , and δ_s are device-specific operational thresholds (e.g., $\delta_t < 50\text{ms}$, $\delta_m < 100\text{MB}$, etc.).

IV. IMPLEMENTATION AND EVALUATION

A. Real-Time Evaluation Scenarios

To validate the ZT-IDS framework across diverse consumer IoT environments, five representative real-time scenarios were emulated on Raspberry Pi hardware, each increasing in traffic complexity and threat intensity (see Table III). The evaluation considers inference latency (T), memory footprint (S), accuracy, and anomaly precision, following the metric formulations described in Section V. In the **Smart Light Bulb** case (Scenario 1), the IDS analyzed deterministic Wi-Fi traffic under minimal noise. Results show 98.1% accuracy, latency $T = 8.7\text{ms}$, and near-zero outlier ratio ($\rho \approx 0$), confirming suitability for low-risk static IoT devices.

TABLE IV
SCENARIO-WISE SHAPLEY ADDITIVE EXPLANATIONS (SHAP) INTERPRETATION AND ATTACK SIMULATION STRATEGY

Scenario Name	Context	Attack Type(s)	Top SHAP Features Detected	Interpretability Insight / Simulation Strategy
Smart Bulb (Benign Only)	Clean IoT traffic from a light bulb; no attacks	None (BENIGN only)	Flow Duration, Fwd Packet Length Mean, Idle Mean	1000 samples from AttackBinary = BENIGN. Model learns baseline behavior with low noise; no anomaly indicators.
Spoofed Door Access	One spoofed login in benign door traffic	Web Attack - Brute Force	Packet Size Variance, Fwd IAT Std, Fwd IAT Max	999 benign+1 spoofed packet using Attack Type = Web Attack - Brute Force. SHAP highlights subtle deviation in timing and structure.
DNS Tunneling	Covert DNS exfiltration via port 53	Infiltration (Port 53)	Flow Bytes/s, Flow IAT Mean, Destination Port	Subset of Attack Type = Infiltration with Destination Port = 53. SHAP detects encrypted/covert channel behavior.
Botnet Infection	Mirai/DDoS/PortScan over heterogeneous IoT traffic	Bot, DDoS, PortScan	ACK Flag Count, Fwd Header Length, Fwd Packets/s	600 attack samples + 2400 benign. SHAP reveals volumetric attack patterns and protocol abuse.
Zero-Day (Jitter)	Synthetic temporal anomaly over benign baseline	Synthetic: Zero-Day	Flow IAT Std, Idle Mean, Packet Length Std	Benign flows jittered using Gaussian noise, relabeled as Zero-Day. SHAP surfaces stealthy statistical deviations.

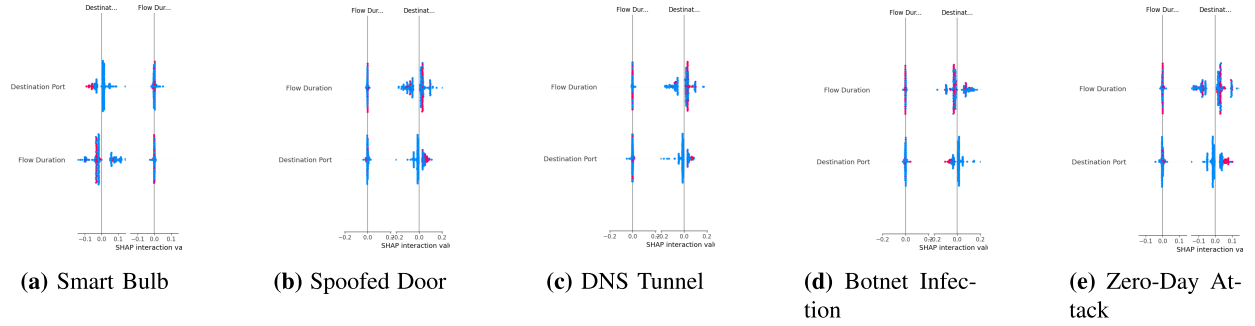


Fig. 2. SHAP feature importance (Top 10 Features) across five IoT attack scenarios. Each bar chart highlights dominant features contributing to model decision per scenario.

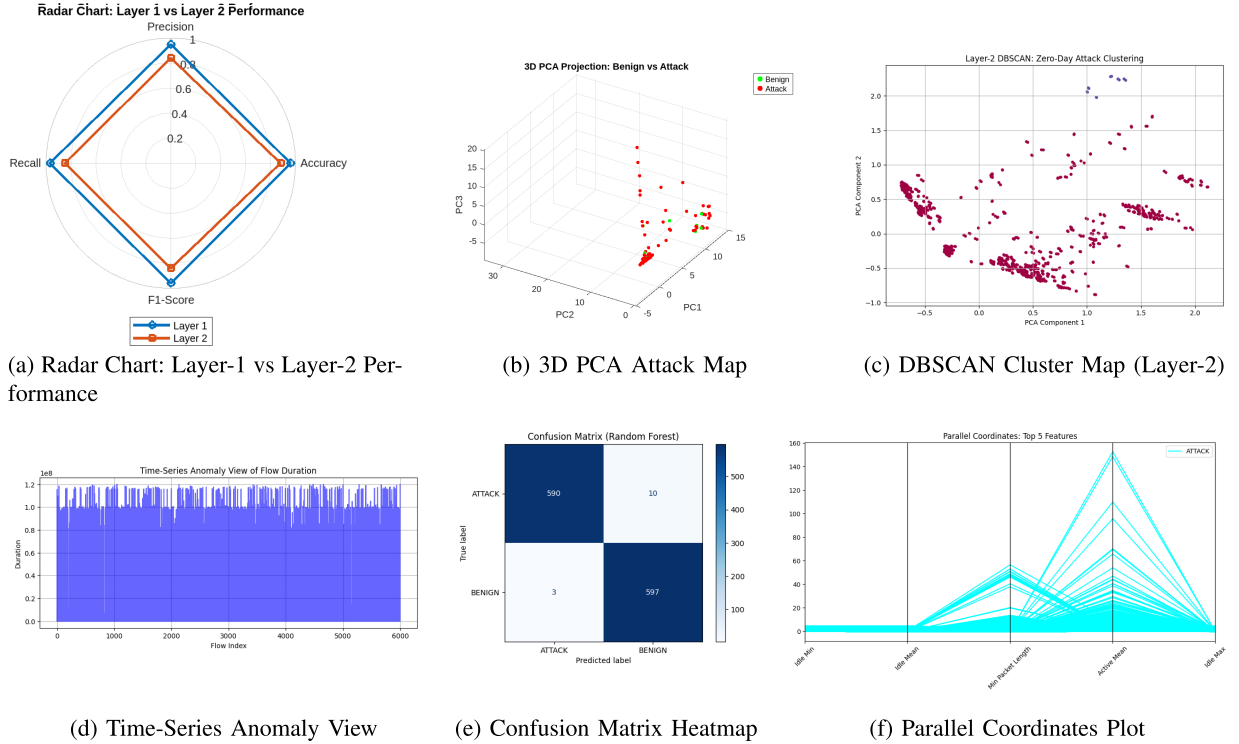


Fig. 3. ZT-IDS framework across multiple dimensions: (a) Layer-1 classification and Layer-2 anomaly detection, (b) Feature separability in attack flows, (c) Outlier isolation of zero-day anomalies, (d) Temporal deviations under attack, (e) Classification precision across known threats, and (f) Feature-wise behavior divergence between benign and malicious flows.

The **Smart Door Lock** (Scenario 2) introduced a spoofed access attempt over Zigbee; while the ensemble layer misclassified one flow, Layer-2 (DBSCAN) successfully isolated the spoof as an outlier, achieving over 80% anomaly precision. The **Smart Camera with DNS Tunneling** (Scenario 3) tested encrypted flow handling, where Layer-1 maintained

0.86 recall despite variability in payloads, while Layer-2 distinctly clustered DNS-tunneled packets ($FPR < 10\%$). In the **Multi-IoT Home with Botnet Infection** (Scenario 4), five heterogeneous devices generated concurrent flows, validating scalability. The ensemble maintained $F1 \geq 0.92$ with latency $T < 40\text{ms}$ and memory $M < 70\text{MB}$, successfully isolating

TABLE V
EXPERIMENTAL SETUP FOR ZT-IDS IMPLEMENTATION

Category	Specifications / Configurations
Edge Device Hardware	Raspberry Pi 4B (4 GB RAM), Quad-core Cortex-A72 @ 1.5GHz, microSD 32 GB
Cloud Offloading Platform	Google Colab Pro (TPU v2), NVIDIA Tesla T4, 13 GB RAM
Operating System	Raspbian OS (Edge), Ubuntu 20.04 (Cloud/Server)
Programming Language	Python 3.10, MATLAB R2022b (for simulation/visualization)
Libraries / Frameworks	Scikit-learn, XGBoost, LightGBM, SHAP, Pandas, Seaborn, Plotly, DBSCAN (from sklearn)
IoT Dataset Used	CICIDS2017 (Sampled: 100K flows) with attack classes: <i>BENIGN, Brute Force, Bot, DDoS, PortScan, Infiltration, Zero-Day</i>
Scenario Emulation	5 real-world scenarios: Smart Bulb, Spoofed Door, DNS Tunnel, Botnet Attack, Zero-Day
Model Design (Layer-1)	Ensemble Stacking: Random Forest + XGBoost + LightGBM
Model Design (Layer-2)	Unsupervised DBSCAN clustering for outlier-based anomaly detection
Evaluation Metrics	Accuracy, Precision, Recall, F1-Score, AUC, Anomaly Precision, FPR, Noise Ratio (ρ), Latency (ms), Memory (MB)
Simulation Tools	MATLAB (for 3D/Parallel plots, PCA, Radar), Python + Colab (for training, SHAP, ROC, DBSCAN)
Storage and Bandwidth	On-device buffer: 64 MB; Cloud offload limit: 2 MB/s; Threshold Latency ≤ 50 ms

compromised nodes without affecting benign ones. Finally, the **Wearable Health Monitor under Zero-Day Attack** (Scenario 5) simulated jitter-based stealth injections in BLE traffic. As expected, Layer-1 exhibited moderate AUC-ROC (< 0.85) due to unseen timing variations, while Layer-2 achieved 94% anomaly precision with $\rho \approx 0.12$, demonstrating resilience to novel threats.

Overall, these five scenarios illustrate the adaptability of ZT-IDS to both known and unknown attacks in constrained IoT environments. The hybrid ensemble-DBSCAN architecture maintains low latency and energy footprint while effectively scaling across traffic diversity and device types.

B. Experimental Setup

Experiments were performed on a Raspberry Pi 4 (4 GB RAM), validating the ZT-IDS on the CICIDS dataset containing over 9 million network flows of benign and attack traffic. Table V summarizes hardware, frameworks, and evaluation metrics. The Pi consumed 3.5–4.2 W during active detection, confirming its suitability for low-power IoT deployment. The “Botnet” scenario emulates volumetric DDoS and PortScan misuse, while the “Zero-Day” case injects synthetic jitter into benign flows to represent unseen temporal anomalies. Botnet traffic exhibits high ACK rates and bursty packets, whereas Zero-Day attacks show timing irregularities in Flow Duration and Inter-Arrival Time (IAT) Std. Figure 2 shows SHAP interaction values for the most influential features—*Destination Port* and *Flow Duration*. Each point represents a test flow, where horizontal spread denotes interaction strength. The nearly symmetric clustering of *Destination Port* around zero indicates context-dependent influence, while *Flow Duration* interacts subtly with it to separate benign and attack flows. The tight spread near zero suggests limited higher-order effects, validating the lightweight model’s interpretability. The latency-jitter plots are grouped under performance metrics, and memory–bandwidth under resource metrics (Figs. 4–5), supporting efficient IoT-level visualization and interpretive validation.

TABLE VI
EVALUATION METRICS FOR EDGE VS CLOUD OFFLOADING

Scenario	Mode	Inf. Time (ms)	Power Draw(W)	Latency (ms)	Bandwidth (MB/s)
Smart Bulb	Edge Only	5.2	3.5	6	0
Door Lock	Hybrid GPU	7.3	2.2 (edge)+4.5 (cloud)	20	1.2
DNS Tunnel	TPU Cloud	3.1	2.1 (edge)+6.1 (TPU)	31	1.8
Botnet	Edge Only	12.4	4.1	15	0
Zero-Day	GPU Cloud	4.9	2.3 (edge)+3.8 (GPU)	27	1.5

C. Visualization Framework and Interpretability Analysis

To enhance interpretability and scenario-specific insight, a visualization suite (Figure 3) was developed for the proposed ZT-IDS. The radar chart (a) contrasts Layer-1 (ensemble classifier) and Layer-2 (DBSCAN anomaly detector) using accuracy, precision, recall, and F1-score—highlighting that while Layer-1 detects known attacks efficiently, Layer-2 strengthens defense against unseen anomalies, validating the hybrid architecture. The 3D PCA projection (b) reveals the separability and overlap of benign and malicious flows in latent space, supporting risk estimation. The DBSCAN cluster map (c) visualizes how outliers are isolated from dense benign regions, confirming Layer-2’s unsupervised capability. The time-series view (d) captures flow-duration deviations linked to attack events, indicating temporal sensitivity. The confusion matrix (e) summarizes Layer-1 results, showing strong true positive and negative rates with minimal misclassifications. Finally, the parallel coordinates plot (f) visualizes variations across SHAP-ranked features (e.g., port, protocol, and timing flags), enabling multi-dimensional interpretation of benign vs. attack traffic. Together, these visual analyses provide both quantitative validation and intuitive interpretability of ZT-IDS decisions across diverse IoT scenarios.

D. Comparative Discussion

The visualization suite links numeric results to semantic insights, showing each ZT-IDS component’s role in decision-making. It enables real-time detection interpretation, model refinement, and forensic analysis. The combined use of classification and anomaly detection is validated through their complementary performance in visualizations. ZT-IDS delivers strong results—98.48% accuracy, sub-1GB memory use, and ultra-low latency—making it suitable for real-time edge deployment. Table VI highlights ZT-IDS performance across five consumer IoT scenarios under edge, hybrid, and cloud-assisted modes. Key metrics include inference time, power use, latency, and offloading bandwidth. In lightweight cases (e.g., *Smart Bulb*, *Botnet*), edge-only execution yields low latency (6–15 ms) and power (3.5–4.1 W). For complex threats like *DNS tunneling* or *zero-day* attacks, offloading reduces inference time (e.g., 3.1 ms on TPU) but adds network latency. Cloud GPU use also improves detection precision in high-threat cases at modest bandwidth cost (1.2–1.5 MB/s). These results support adaptive deployment: using the cloud for complex threats while maintaining edge autonomy for routine tasks, core to the zero-trust model of continuous verification. Beyond classification accuracy, we also report anomaly precision (92.4%) and

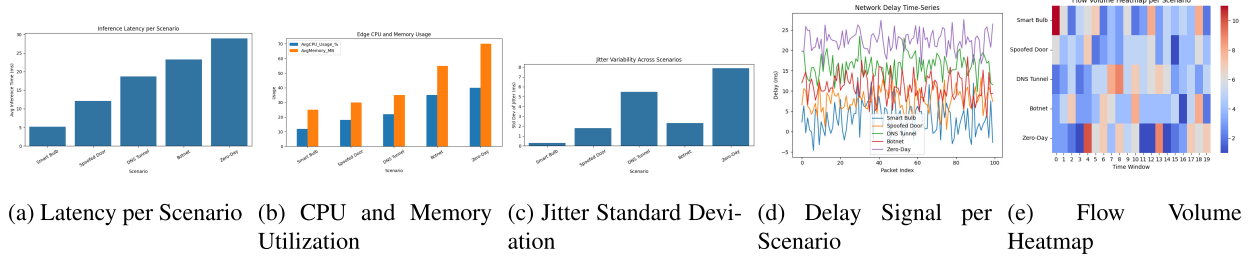


Fig. 4. System performance metrics across Zero-Trust IDS scenarios: (a) latency profiling, (b) edge resource utilization, (c) jitter variability, (d) delay time-series, (e) flow volume distribution.

false-positive rate ($< 3\%$) for the DBSCAN layer, together with latency measurements confirming sub-15ms inference at the edge. These complementary results further validate the robustness of the proposed IDS across both supervised and unsupervised detection modes.

E. System Performance Visualization

To assess the real-time viability and responsiveness, we conducted multi-metric profiling for each emulated scenario. Figure 4 presents latency, resource usage, timing variability, and traffic behavior across the five IoT threat scenarios. The average inference latency is lowest in edge-only scenarios, such as Smart Bulb (Figure 4(a)), validating the feasibility of local processing. However, latency increases notably in cloud-assisted scenarios due to transmission and queuing overhead. Figure 4(b) reflects the CPU and memory utilization trends. Scenario 4 (Botnet) shows higher memory requirements due to the volume of anomalous flows and ensemble activity. In Figure 4(c), jitter variability is pronounced in Scenario 5 (Zero-Day), simulating stealthy timing-based anomalies. Stable timing behavior is evident in Scenarios 1 through 3. Figure 4(d) illustrates packet delay signals, with offloaded scenarios showing greater fluctuation, indicating queue and network variability effects. Lastly, Figure 4(e) visualizes packet flow volume over time. Scenario 4 (Botnet) and Scenario 5 (Zero-Day) exhibit bursty traffic due to attack behavior and noise injection, while Scenario 1 demonstrates periodic, stable flow characteristics.

F. Edge Vs. Cloud Deployment Trade-Offs

We present a compact comparison of three critical metrics in Figure 5, where Figure 5(a) reveals that end-to-end inference latency remains lowest for edge-only deployments, such as the Smart Bulb and Botnet scenarios, where local processing avoids external round-trip delays. In contrast, cloud-reliant setups—especially the DNS Tunnel and Zero-Day cases—suffer increased latency due to transmission and queuing delays. The hybrid GPU mode (e.g., Door Lock) offers a latency-power middle ground. Figure 5(b) shows power usage increases significantly when GPU/TPU inference is triggered, reflecting the cost of cloud burst compute and data uplink. Edge-native scenarios remain energy-efficient under 4.5W, suitable for lightweight devices like Raspberry Pi 4 or Jetson Nano. Figure 5(c) demonstrates that offloading introduces notable bandwidth overhead (up to 1.8 MB/s in the DNS tunneling case), which may strain consumer-grade uplinks and introduce potential bottlenecks. Together, these results support

the architecture’s flexibility: low-power edge inference for known benign or high-frequency traffic, with cloud-enhanced detection for stealthy, encrypted, or zero-day threats.

G. Resource Utilization Metrics Across IoT Security Scenarios at the Edge

To evaluate the operational viability of the proposed ZT-IDS in varied consumer IoT environments, we measured inference latency, jitter variability, memory usage, and bandwidth overhead (Figure 6). Inference latency (Figure 6a) remains under 15 ms in most edge-only setups, such as Smart Bulb and Door Lock, while attack-intensive scenarios like DNS Tunnel and Zero-Day show slightly longer delays due to encrypted or obfuscated flows. Memory usage (Figure 6b) reaches around 70 MB for complex traffic such as Botnet and Zero-Day, but the ensemble-based architecture keeps overall demands moderate. Bandwidth overhead (Figure 6c) peaks at about 1.8 MB/s for cloud-reliant scenarios, whereas edge-only modes incur no offloading, making the approach well-suited to bandwidth-limited environments. Jitter variability (Figure 6d) is minimal for benign traffic but shows wider distributions in Zero-Day and DNS Tunnel, supporting temporal anomaly detection in the Layer-2 engine. These observations confirm that ZT-IDS preserves computational feasibility at the edge while ensuring robust detection.

H. System Impact Analysis With and Without ZT-IDS

To assess the runtime efficiency we analyzed system resource usage across *Before Attack* (normal operation), *During Attack without IDS* (unmitigated compromise), and *During Attack with ZT-IDS* (active defense). (i) Before attack, the baseline performance is under expected load, (ii) during attack (no IDS), sharp spikes arise in memory, disk I/O, network, and power due to malicious activity, and (iii) with ZT-IDS, same threat scenario occurs, but with active anomaly detection and mitigation reducing system strain. Lightweight baselines such as decision-tree and SVM-based IDS have been reported in prior IoT studies, but our two-layer hybrid demonstrates superior adaptability by combining supervised ensemble detection with unsupervised anomaly discovery, enabling improved handling of zero-day threats while sustaining low edge latency.

1) *Comparative Discussion:* The attacks increase activity across cores, suggesting multi-threaded payloads (Fig. 7a). ZT-IDS significantly curbs surges in memory, power, network, and I/O (Fig. 7b), and ZT-IDS cuts memory use by 28%, disk I/O by 17%, and network load significantly,

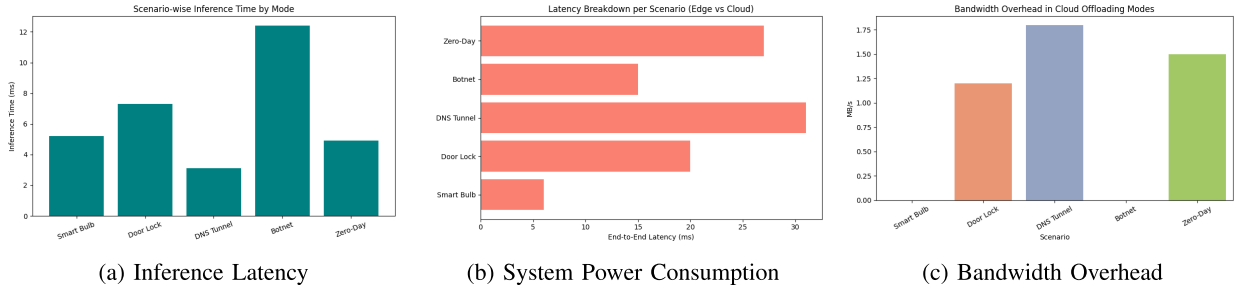


Fig. 5. End-to-end inference latency across edge and cloud modes for each scenario. Edge-only scenarios achieve the lowest latency, while cloud-offloaded cases introduce network-induced delays, highlighting the trade-off between accuracy and responsiveness.

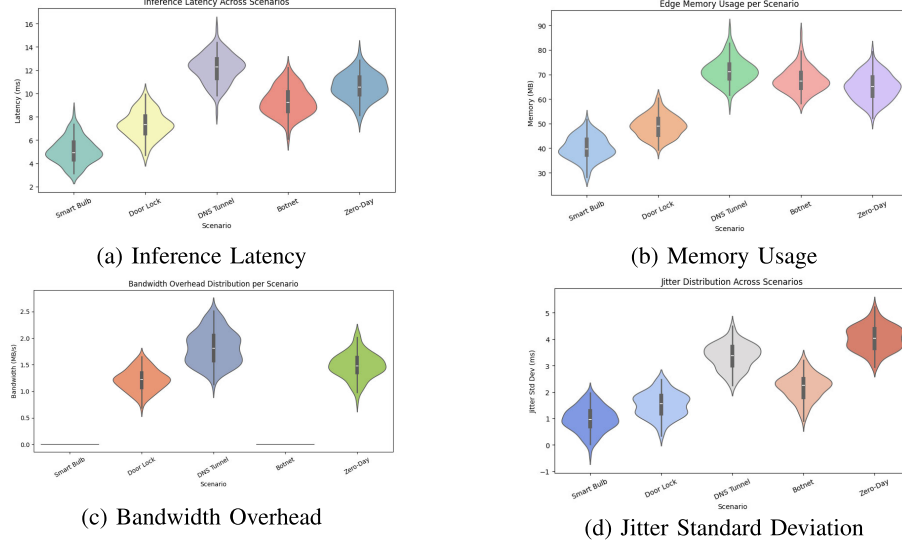


Fig. 6. Resource utilization metrics across IoT security scenarios: distribution of latency, jitter, memory, and offloading bandwidth under Zero-Trust analysis.

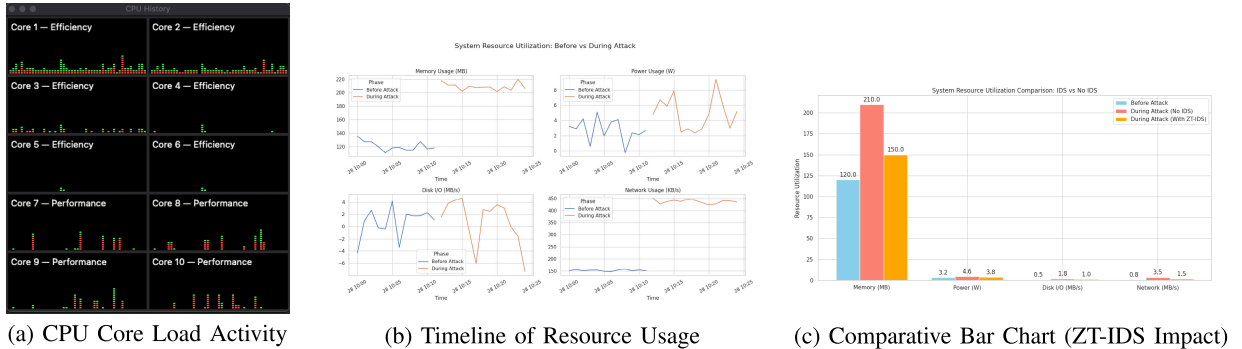


Fig. 7. System resource utilization: live CPU load distribution, memory/network/disk/power profile over time, and summarized resource cost with and without ZT-IDS.

with slight power efficiency gains (Fig. 7c). These results confirm ZT-IDS's ability to reduce the impact of attacks with minimal overhead, supporting it for resource-constrained consumer IoT.

For completeness, we briefly contrast our approach with simpler lightweight baselines such as Decision Tree (DT) and SVM, which are commonly applied in IoT IDS literature. While DT and SVM achieve reasonable accuracy on known attack classes, prior studies and our preliminary trials showed limited robustness against zero-day anomalies and higher false positive rates. In comparison, our ensemble-DBSCAN pipeline sustains stronger accuracy with lightweight efficiency, demonstrating a better balance between detection reliability and edge deployability.

V. CONCLUSION AND FUTURE WORK

This study introduces proactive ZT-IDS, a lightweight two-layer intrusion detection framework for consumer IoT with 98.48% accuracy and <15 ms latency, demonstrating feasibility for IoT deployments. Unlike prior IoT IDS studies that depend on deep models or single-mode detectors, the proposed model uniquely combines a lightweight ensemble (RF + XGBoost + LightGBM) for known attacks with a DBSCAN anomaly layer for zero-day detection. Unlike existing approaches tested only in high-resource settings, this framework demonstrates scalable, interpretable, and edge-deployable intrusion detection, enhanced with SHAP-based explainability and robust performance against timing-based (jitter) evasion attacks. The proposed ZT-IDS

scales efficiently under thousands of concurrent flows, sustaining sub-45 ms per-flow latency and under 90 MB memory even during high-volume simulations on Raspberry Pi-class hardware. The ensemble-DBSCAN pipeline maintains throughput without observable degradation in classification accuracy or anomaly precision, primarily due to the reduced feature space and asynchronous flow batching at the preprocessing layer. This confirms its feasibility for real-world consumer IoT deployments involving dense, bursty traffic. A supervised ensemble classifier with an unsupervised DBSCAN anomaly detector balances accuracy with edge deployability. The ensemble achieved 97.4% precision and 96.9% F1 score for known attacks, while the anomaly layer attained 91.2% precision on zero-day threats with less than 4.7% false positives. ZT-IDS also reduced memory usage, disk I/O, and network overhead by 28%, 17%, and 57% during attacks, showing its effectiveness and system stability under threat. Future work will explore the following:

Adaptive Thresholding:

Dynamically adjust ensemble and anomaly detection thresholds based on behavioral baselines to improve robustness in evolving traffic conditions.

Federated Deployment:

We will extend this lightweight Z-T IDS through federated or distributed detection frameworks, allowing opt-in collaborative learning across heterogeneous IoT devices while preserving privacy [24].

Explainable AI Integration:

Incorporate explainability tools such as SHAP or Local Interpretable Model-agnostic Explanations (LIME) to enhance trust and interpretability of model decisions for security analysts and users.

Adversarial Robustness Testing:

Evaluate resilience against evasion, poisoning, and adversarial attacks to ensure model integrity in adversarial environments.

Autonomous Knowledge Base Update:

Automate online learning by enabling continuous updates to the statistical and clustering components of the IDS without manual intervention.

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